

Machine Learning Model Accuracy Report

ALGORITHM EVALUATION AND METRICS REPORT

Model Designation: Fraud-Detection-XGB-v2.1 Evaluation Date: August 18, 2026 Lead Data Scientist: Dr. Arthur Pendelton

Overview: This report summarizes the offline evaluation metrics for the newly trained XGBoost classification model (v2.1), designed to flag fraudulent credit card transactions in real-time. The model was trained on a historical dataset of 12 million transactions from Q1 and Q2 2026, featuring 85 engineered features including geospatial velocity, transaction velocity, and behavioral anomaly scoring.

Evaluation Methodology: The dataset was split using a stratified 80/20 train/test split to preserve the extreme class imbalance (fraudulent transactions represent only 0.15% of the total dataset). Evaluation was performed exclusively on the 20% holdout set. The primary optimization metric for this deployment is the F1-Score, aiming to balance precision (minimizing false positives that frustrate customers) and recall (catching true fraudulent behavior).

Key Metrics: - Accuracy: 99.91% (Note: Misleading due to severe class imbalance) - Precision: 0.88 (88% of transactions flagged as fraud were actually fraud) - Recall: 0.81 (The model successfully identified 81% of all true fraudulent transactions) - F1-Score: 0.843 - ROC-AUC: 0.965

Feature Importance (Top 3): 1. Distance from primary billing zip code (SHAP value: +3.2) 2. Time since last transaction (SHAP value: -2.1) 3. Transaction size relative to 30-day moving average (SHAP value: +1.8)

Business Impact & Conclusion: Model v2.1 demonstrates a 14% improvement in Recall compared to the incumbent Random Forest model currently in production, without sacrificing Precision. This translates to an estimated \$450,000 in monthly fraud prevention savings.

We recommend deploying v2.1 as a "shadow model" alongside the production environment for 14 days. If the real-time drift metrics remain stable, it should be promoted to the primary inference endpoint by early September.